

Health-State Transitions and the Lifetime Cost of Aging: A Multi-State Actuarial Model of Medicare Spending and Household Out-of-Pocket Tail Risk

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Abstract

Medicare's Hospital Insurance trust fund is projected to be depleted in 2033, yet aggregate projections say little about how the cost of aging falls on individual households. We build a five-state continuous-time Markov model of later-life health (healthy, chronic illness, disability, long-term-care need, dead) from 213,552 intervals between Health and Retirement Study interviews, 1998 to 2022, with deaths dated to the month. State-specific Medicare costs come from the Medicare Current Beneficiary Survey and out-of-pocket cost distributions from HRS core and exit interviews. Mortality is calibrated to the 2023 US life table, and lifetime costs from age 65 are projected by microsimulation and checked against an exact recursion. For a 65-year-old drawn from the population, the present value at 3% of lifetime Medicare spending is \$153,316 for men and \$172,481 for women, and of out-of-pocket spending on care \$37,660 and \$48,975, before Part B premiums. The distribution is strongly skewed: the mean of the worst 5% of outcomes (CVaR95) is \$127,541 for men and \$174,866 for women, and 77.8% and 88.3% of lives in that tail pass through long-term-care need, against 46.4% and 60.6% of all lives. Health at 65 changes the timing of costs far more than their expected value. Passing half of the post-2033 Part A shortfall to beneficiaries raises expected lifetime out-of-pocket cost by \$872 for men and \$1,042 for women; real cost growth at the Trustees' Part B rate raises CVaR95 by \$36,990 and \$55,219. Code and a public calculator are released.

Keywords. multi-state model; interval-censored panel data; Medicare; long-term care; out-of-pocket spending; tail risk; microsimulation; Health and Retirement Study

1. Introduction

The 2026 Medicare Trustees Report projects that the Hospital Insurance (HI) trust fund will be depleted in the second quarter of 2033, after which incoming revenue would cover 89% of scheduled Part A benefits, and that Medicare spending will rise from 3.9% of GDP in 2025 to 6.5% in 2050 (Boards of Trustees, 2026). These projections are aggregates. They answer what the program will cost, not what aging will cost a particular household, how uncertain that cost is, or which health paths produce the bad outcomes. Those questions matter to the people who price and regulate long-term-care, Medigap and Medicare Advantage products, to planners advising retirees, and to policymakers choosing how a financing gap is closed.

The actuarial tool for this problem is the multi-state model. A person moves among health states with age-dependent intensities; each state carries a cost; the expected present value of lifetime cost follows from the transition probabilities, and its distribution follows from simulating paths. Multi-state models are standard in disability and long-term-care insurance (Haberman and Pitacco, 2018) and have been estimated on the Health and Retirement Study (HRS) for functional disability (Fong, Shao and Sherris, 2015; Sherris and Wei, 2021). Economists, separately, have measured medical-expense risk in retirement and its consequences for saving

(Hubbard, Skinner and Zeldes, 1995; De Nardi, French and Jones, 2010; Jones et al., 2018). The two strands rarely meet in one model that is payer-specific, reports the tail, and connects to Medicare's financing.

This paper builds that model with public data and released code. We ask four questions.

- RQ1. What are the transition intensities among healthy (H), chronic illness (C), disability (D), long-term-care need (L) and death (X) for Americans aged 50 and over, and how do they vary with age, sex, education and race-ethnicity?
- RQ2. What are expected lifetime Medicare and household out-of-pocket costs from 65, by health state at 65, sex and income?
- RQ3. What is the distribution of the present value of lifetime out-of-pocket cost, what are its Value-at-Risk and Conditional Value-at-Risk, and which paths produce the tail?
- RQ4. How would plausible responses to the post-2033 HI shortfall change household out-of-pocket tail risk and asset exhaustion?

Four design choices shape the answers. The transition model is fitted to interval-censored panel data by exact likelihood, so transitions between biennial interviews are not assumed away. The model's mortality is tested against the 2023 US life table and then calibrated to it, keeping the state relativities the data identify. Out-of-pocket costs are drawn from full empirical distributions by state, not means, and the last year of life is drawn from exit interviews, which core interviews never observe. And the expected values reported by simulation are checked against an exact recursion.

The main results are as follows. Life expectancy at 65 is 18.3 years for men and 20.8 for women, of which 1.58 and 2.60 years are spent in long-term-care need; 46.4% of men and 60.6% of women enter that state. Lifetime Medicare spending varies little with health at 65, because poorer health shortens the years over which costs accrue. Out-of-pocket risk is concentrated: CVaR95 is 3.4 times the mean for men and 3.6 times for women, and the tail is dominated by long-term-care paths. A partial pass-through of the HI shortfall moves expected out-of-pocket cost by about 2%, small beside the effect of faster real cost growth, which falls mostly on the tail.

2. Background

Medical-expense risk in retirement. Uncertain medical costs affect saving and asset decumulation (Hubbard, Skinner and Zeldes, 1995; Palumbo, 1999), and De Nardi, French and Jones (2010) showed that medical expenses that rise with age and income help explain why the elderly run down assets slowly. De Nardi et al. (2016) documented medical spending of the US elderly by payer, and Jones et al. (2018) estimated lifetime medical spending of retirees with a dynamic model of health and spending. End-of-life spending has been measured from HRS exit interviews (Marshall, McGarry and Skinner, 2011; Kelley et al., 2013) and is highest for people with dementia (Kelley et al., 2015; Hurd et al., 2013). Hurd, Michaud and Rohwedder (2017) showed that lifetime nursing-home use and its out-of-pocket cost are concentrated in a minority of people. Medicaid's role as payer of last resort shapes demand for private long-term-care insurance (Brown and Finkelstein, 2008), and preferences over long-term care shape late-life saving (Ameriks et al., 2020).

Health paths and cumulative spending. Lubitz et al. (2003) found that older people in better health live longer but accumulate similar Medicare spending over their remaining lives. Multi-state health expectancies go back at least to Crimmins, Hayward and Saito (1994), and microsimulation of the future elderly is the basis of the Future Elderly Model (Goldman et al., 2005).

Actuarial multi-state models. Haberman and Pitacco (2018) set out the multiple-state framework for disability and long-term-care insurance. Fong, Shao and Sherris (2015) estimated multi-state models of functional disability from HRS; Shao, Sherris and Fong (2017) used such a model for long-term-care insurance pricing and solvency capital; Li, Shao and Sherris (2017) and Sherris and Wei (2021) added systematic trend and uncertainty.

Estimation from panel data. When states are observed only at interviews, the likelihood of a continuous-time Markov model is built from transition probabilities over the observed intervals (Kalbfleisch and Lawless, 1985), as implemented in the `msm` package (Jackson, 2011) and treated at length by van den Hout (2016). Titman and Sharples (2010) review diagnostics. Competing-risk and multi-state models for exactly observed transitions are covered by Putter, Fiocco and Geskus (2007) and de Wreede, Fiocco and Putter (2011).

What this paper adds. It combines a multi-state health model estimated on interval-censored HRS data with payer-specific costs from the MCBS, empirical out-of-pocket distributions including end-of-life costs, tail measures (Rockafellar and Uryasev, 2000), and scenarios tied to the Trustees' financing projections, and it releases the whole pipeline and a public calculator.

3. Data

HRS panel. We use the RAND HRS Longitudinal File 2022 (V1) (Bugliari et al., 2025; Sonnega et al., 2014), waves 4 to 16 (1998 to 2022), respondents aged 50 and over at interview. Each interview is assigned a state:

- H: no ever-diagnosed condition among eight (hypertension, diabetes, cancer, lung disease, heart problems, stroke, psychiatric problems, arthritis) and no limitation in the five RAND activities of daily living (ADL);
- C: at least one condition and no ADL limitation;
- D: one or two ADL limitations;
- L: three or more ADL limitations, or living in a nursing home at interview;
- X: dead, dated to the month from the RAND death date.

The panel has 40,354 respondents with 237,944 classified person-interviews (Table 1). Consecutive interviews give 197,590 intervals with a mean length of 2.0 years, and 15,962 deaths are dated exactly; persons last seen alive are censored at that interview. Weighted, 16.5% of person-interviews are in H, 68.6% in C, 10.8% in D and 4.1% in L. Observed transitions are in Table A1. Because conditions are ever diagnosed, C to H does not occur and is not modeled.

Out-of-pocket spending. Core interviews report out-of-pocket medical spending over the previous two years, excluding premiums, with RAND imputations. We halve it to an annual figure, express it in 2024 dollars with the CPI-U, and attach it to the state observed at that interview. The exit interview, answered by a proxy after death, reports out-of-pocket spending from the last core interview to death. We take the final-year amount as that spending times $\min(1, 12/\text{months})$, which assigns a longer window's spending evenly and so understates the final year; 14,576 decedents aged 65 and over contribute.

Medicare spending. The MCBS Cost Supplement public use files for 2019 and 2021 to 2023 (29,144 beneficiary-years) give annual spending by payer, with Medicare Advantage payments included, in cells of age by count of chronic conditions (0-1, 2-3, 4+). Standard errors use the survey's balanced repeated replication with Fay's adjustment of 0.30. The pooled Medicare mean is \$11,802 (standard error \$278), 44.2% of beneficiary-years are in Medicare Advantage, and inpatient plus home health account for 17.6% of all-payer spending (Table A6). The Cost Supplement excludes facility, hospice and institutional events, which fall mostly in the D and L states.

Other sources. Mortality benchmarks are the 2023 US period life tables (Arias and Xu, 2025), selected ages in Table A7. HI depletion timing, the payable share and cost growth rates are from the 2026 Trustees Report. The standard Part B premium is \$174.70 a month in 2024 (CMS, 2023).

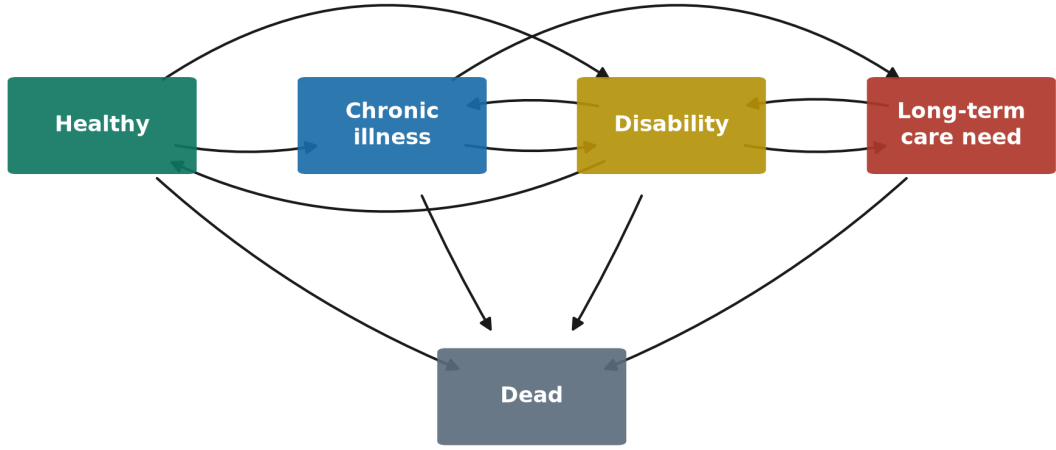
4. Model

4.1 Transition intensities

Let $S = \{H, C, D, L, X\}$ and let $q_{jk}(x, z)$ be the intensity of moving from j to k at age x for covariates z . Twelve transitions are allowed (Figure 1): H to C, D, X; C to D, L, X; D to H, C, L, X; L to D, X. Each intensity is log-linear,

$$\log q_{jk}(x, z) = \beta_{0,jk} + \beta_{1,jk} \frac{x - 65}{10} + \gamma_{jk}^\top z,$$

a Gompertz form in age, with $z = (\text{female, any college, nonwhite or Hispanic})$. The intensity matrix $Q(x, z)$ has off-diagonal entries q_{jk} and rows summing to zero. Over an interval of length d starting at age x , Q is held at the midpoint age, so $P(x, x + d) = \exp\{Q(x + d/2, z)d\}$, the solution of the Kolmogorov forward equation for piecewise-constant intensities.



Conditions are ever diagnosed, so Chronic illness does not return to Healthy. Twelve intensities are estimated.

Figure 1. State space and allowed transitions.

For person i observed in state r at age x and in state s after d years, the likelihood contribution is $P_{rs}(x, x + d)$. For a death at time d after the last interview in state r , the state just before death is unobserved and the contribution is

$$\sum_{k \neq X} P_{rk}(x, x + d) q_{kX}(x + d, z).$$

The log likelihood is maximized with gradients from automatic differentiation through the matrix exponential; standard errors come from the inverse observed information. Estimation is unweighted, the usual convention for panel multi-state models, with sex, education and race-ethnicity entering as covariates; weighted prevalence is used for validation. Starting values come from crude rates. The fitted model is checked against a simulated panel with known intensities in the released tests.

4.2 Population at 65 and mortality calibration

A simulated 65-year-old is drawn from the weighted joint distribution of state, education and race-ethnicity among HRS respondents aged 64 to 66, with nursing-home residents included through their own weight. For men the state mix is 14.6% H, 73.5% C, 8.9% D and 3.0% L; for women 14.4%, 72.3%, 9.2% and 4.0%.

The fitted model pools deaths from 1998 to 2022. Its life expectancy at 65 for the entry mix is 16.72 years for men and 19.54 for women, against 18.19 and 20.71 in the 2023 table (Table 3). Rather than change the relative mortality of the states, which the data identify, we scale every death intensity by

$$\lambda(x) = \exp\{c_0 + c_1(x - 65)/10\},$$

by sex, with (c_0, c_1) minimizing the l_x -weighted squared difference between the model's and the table's $\log q_x$ at ages 65 to 99. The multiplier runs from 0.70 at 65 to 0.92 at 95 for men and from 0.64 to 1.05 for women. The uncalibrated model is kept as a robustness variant.

4.3 State costs

Medicare cost in state j and age band b (65-74, 75+) is the MCBS cell mean for band b averaged over the distribution of chronic-condition counts among HRS respondents in j and b . The HRS counts eight conditions and the MCBS a longer list, so this mapping places HRS respondents in lighter cells than the Medicare population they represent: the cohort's implied Medicare spending per person-year is 0.846 and 0.866 of the MCBS mean in the two bands (Table A4). We keep the state relativities and scale the level in each band so the weighted HRS average equals the MCBS mean, factors of 1.20 and 1.17. After scaling, the model cohort reproduces the MCBS means to within 1.6%.

Out-of-pocket cost in a year alive in state j is drawn from the weighted empirical HRS distribution for j , the age band and sex. In the year of death the draw is replaced by one from the exit-interview distribution for the age band and sex. The standard Part B premium is added for each year alive and reported separately.

4.4 Lifetime cost and its distribution

Lives step through one-year transition matrices $P(x, x+1)$ from 65 to 110. With costs accruing at the start of each year and discount factor $v = 1/1.03$, the present value of out-of-pocket cost for a life with states J_t at age $65 + t$ is

$$PV^O = \sum_{t=0}^{44} v^t \left[\mathbb{1}\{J_t \neq X\} O_t + \mathbb{1}\{J_t \neq X, J_{t+1} = X\} (E_t - O_t) \right],$$

where O_t is the ordinary draw and E_t the end-of-life draw; PV^M and the premium are defined in the same way. With $\pi(t)$ the state distribution at $65 + t$, $\pi(t+1) = \pi(t)P(65+t, 66+t)$, and μ_j and μ^E the means of the draws, the expected values have the closed form

$$\mathbb{E} PV^O = \sum_t v^t \sum_{j \neq X} \pi_j(t) [\mu_j + P_{jX}(65+t, 66+t) (\mu^E - \mu_j)],$$

which the simulation means must match. Tail risk is summarized by VaR_α , the α quantile of PV^O , and $\text{CVaR}_\alpha = \mathbb{E}[PV^O \mid PV^O \geq \text{VaR}_\alpha]$ at $\alpha = 0.90, 0.95$ and 0.99 (Rockafellar and Uryasev, 2000). We simulate 100,000 lives for each sex and each health state at 65, and 100,000 from the population mix. Parameter uncertainty is the spread of the exact expected values over 200 draws of the intensity parameters from their asymptotic normal distribution.

For income, household income is ranked into weighted tertiles within interview wave and age group. A life in tertile k starts from the entry mix of tertile k at 64 to 66 and draws out-of-pocket costs from tertile k at every age.

4.5 Financing scenarios

The scenarios apply to a cohort turning 65 in 2026, so calendar year is 2026 plus years since 65. Part A's share of each state's Medicare cost is the MCBS inpatient-plus-home-health share within each chronic-condition cell, weighted by the state's condition mix.

- S0: benefits paid in full, the Trustees' convention.
- S1: from 2033 Part A pays 89% of scheduled benefits, and a share s of the shortfall becomes out-of-pocket cost, for $s = 0$ (a pure provider payment cut), 25%, 50% and 100%.
- S2: the whole cohort shortfall falls on person-years in long-term-care need, as a service-specific cut to post-acute coverage would.
- S3: S1 with $s = 50\%$ plus real cost growth at the Trustees' Part B per-beneficiary rate of 1.7% a year.

Asset exhaustion compares cumulative undiscounted out-of-pocket cost with household net assets at 65 by income tertile, with no investment returns or income. It measures exposure, the point at which Medicaid eligibility would come into view, not a forecast of Medicaid enrollment.

5. Validation

Mortality. Calibration brings life expectancy at 65 to 18.31 years for men and 20.87 for women, within 0.16 years of the table (Table 3). One-year death probabilities match closely from 70 to 85. The log-linear multiplier cannot follow the table's steep rise past 90: at 95 the calibrated model gives 0.1939 for men against 0.2563 in the table, so survival at the oldest ages, and costs there, are somewhat overstated.

Prevalence. Figure 2 and Table A3 compare the calibrated model's state distribution among survivors with the weighted HRS cross-section. Agreement is within 3 percentage points in every state up to age 80 for both sexes. The model understates H at older ages, because it follows a cohort from 65 while the cross-section includes people interviewed with missing condition histories, and understates L among women over 90 (29.1% against 34.4%).

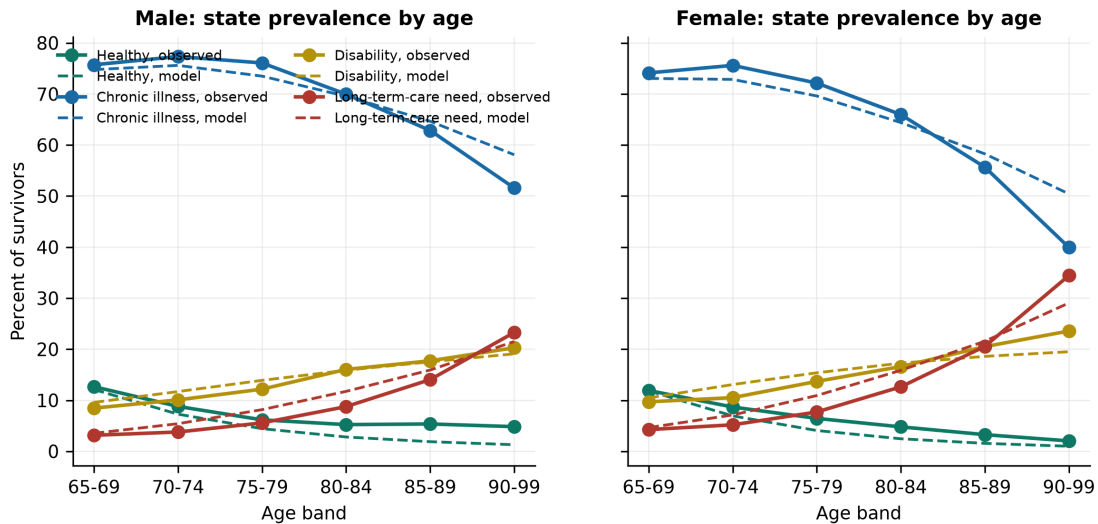


Figure 2. State prevalence by age: weighted HRS cross-section (solid) and calibrated model (dashed).

Spending. After the level scaling in Section 4.3 the model's Medicare spending per person-year is 1.013 and 1.016 times the MCBS mean in the two age bands (Table A4). The simulation means agree with the exact recursion: population-mix lifetime out-of-pocket cost is \$37,660 simulated against \$37,703 exact for men.

6. Results

6.1 Transitions (RQ1)

Table 2 reports the intensities; the likelihood-ratio test for the education and race-ethnicity terms is 1,512.4 on 24 degrees of freedom. Death intensities rise with age in every state, by a factor of 2.35 per decade from H and 2.07 from C, and more slowly from D (1.42) and L (1.54), the flattening expected when frailer people leave the at-risk group first. Women die at 0.42 to 0.68 of men's rate from each state, and are 12% more likely than men to move from C to D. Any college lowers the rate of moving from C to D (hazard ratio 0.75) and from C to L (0.40), and lowers mortality from H (0.46) and C (0.80). Being nonwhite or Hispanic raises the rate from C to D (1.34) and from C to L (2.35), while mortality from D and L is lower (0.64 and 0.76), a pattern consistent with selection into those states. The direct C to L intensity falls with age (0.16 per decade, interval 0.12 to 0.22): with two-year intervals, the model attributes most moves from C to L at older ages to a path through D, whose intensities rise with age. One-year probabilities are in Table A2 and Figure 3.

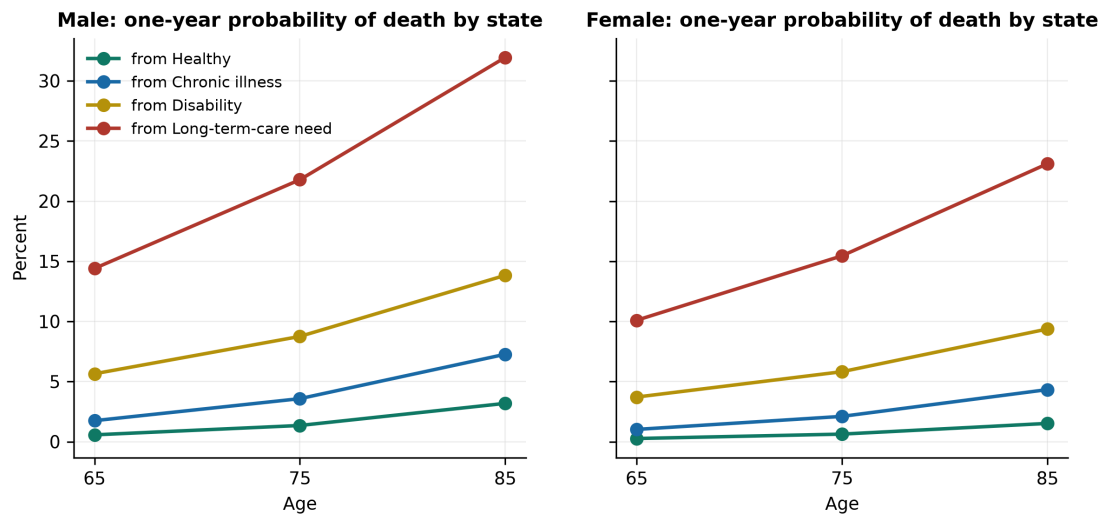


Figure 3. One-year probability of death by state at ages 65, 75 and 85, fitted model, reference covariates.

6.2 Health expectancies

A 65-year-old man drawn from the population can expect 18.3 more years: 1.15 healthy, 13.15 with chronic illness, 2.45 with disability and 1.58 in long-term-care need (Table 5b and Figure 4). A woman can expect 20.8 years, with 3.09 in disability and 2.60 in long-term-care need. Starting healthy adds about two years of life for either sex against the population mix; starting in L cuts life expectancy to 12.5 years for men and 15.7 for women. By covariate profile before calibration (Table A5), education moves life expectancy at 65 by more than race-ethnicity does.

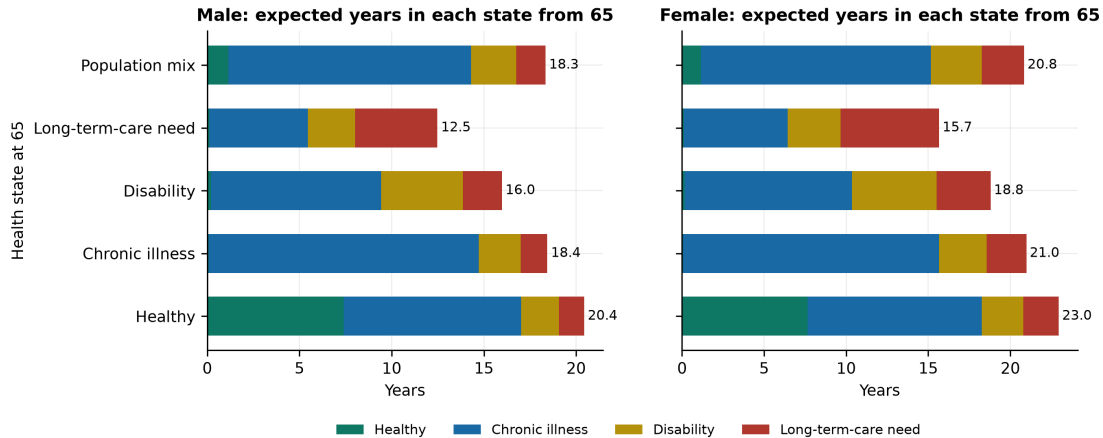


Figure 4. Expected years in each state from 65, by health state at 65.

6.3 Annual costs by state

Medicare spending per year rises from \$5,291 in H to \$14,048 in L at ages 65 to 74, and from \$7,010 to \$14,527 at 75 and over (Table 4). Out-of-pocket spending is where the states differ most. Mean annual spending is about \$1,200 in H, \$2,000 to \$2,300 in C and \$2,600 to \$3,100 in D, but \$9,228 for men and \$10,829 for women in L at 75 and over, with a 99th percentile of \$101,341 and \$122,600 (Table 4b). Out-of-pocket spending in the last year of life averages \$9,913 for men and \$12,641 for women who die at 75 or older, and its 99th percentile is \$125,678 and \$151,655 (Table 4c).

6.4 Expected lifetime cost (RQ2)

Table 5 gives expected lifetime costs from 65. For the population mix the present value of Medicare spending is \$153,316 for men (95% parameter interval \$151,948 to \$154,224) and \$172,481 for women. Out-of-pocket spending on care is \$37,660 for men (interval \$37,398 to \$38,002) and \$48,975 for women, with medians of \$31,282 and \$38,684. The Part B premium adds \$29,224 and \$32,221. The end-of-life step contributes \$2,663 and \$3,072 of the out-of-pocket total. Parameter uncertainty is small relative to the spread across lives: the intervals for expected values are within about 1% of the mean.

Health at 65 changes expected lifetime Medicare cost less than it changes annual cost. A man who is healthy at 65 is expected to cost Medicare \$140,295 over his life; one in long-term-care need, whose annual costs are more than twice as high at 65, is expected to cost \$126,204, because he is expected to live 12.5 years rather than 20.4. The highest expected cost is for men and women with chronic illness but no limitation at 65, \$157,627 and \$176,905, who combine long lives with high annual cost. Expected out-of-pocket cost varies even less: \$35,373 for men healthy at 65 and \$38,065 for men in L.

Income changes the composition more than the total. Low-income 65-year-olds are more often in L at entry (6.8% of men and 8.0% of women in the lowest tertile, against 0.8% in the highest) and more often reach it (51.1% of men against 43.6%), but live shorter lives and draw on lower out-of-pocket spending in each state, partly because Medicaid pays. Their expected lifetime out-of-pocket cost is \$30,387 for men and \$42,713 for women, against \$44,277 and \$59,647 in the highest tertile (Table 5c). Because the transition model has no income term, the life-expectancy gap by income (17.4 against 19.1 years for men) is understated.

6.5 The tail (RQ3)

Figure 5 shows the distribution of lifetime out-of-pocket cost for the population mix. For men, VaR95 is \$92,761 and CVaR95 is \$127,541, 3.4 times the mean; CVaR99 is \$188,051. For women, VaR95 is \$122,555, CVaR95 \$174,866 and CVaR99 \$282,008 (Table 6).

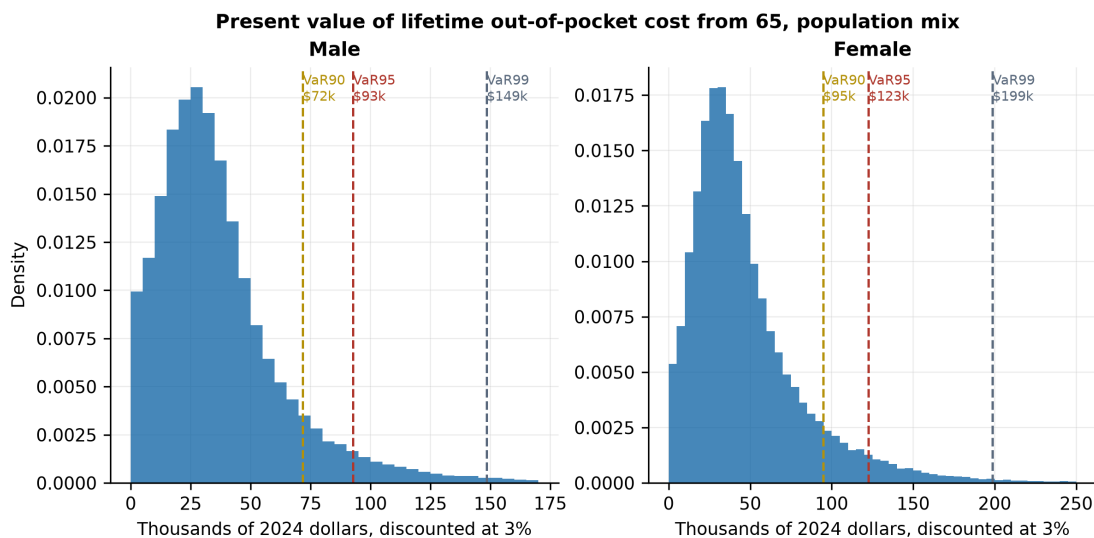


Figure 5. Distribution of the present value of lifetime out-of-pocket cost from 65, population mix, with Value-at-Risk at 90%, 95% and 99%.

The tail is a long-term-care phenomenon. Among men in the worst 5% of outcomes, 77.8% passed through L, against 46.4% of all men; for women 88.3% against 60.6%. People in the tail also live longer, dying at 87.4 on average against 83.3 for all men, since a long stay in L requires survival. The end-of-life draw accounts for 19.8% of the tail's present value for men and 16.1% for women. Starting in L raises CVaR95 to \$158,274 for men and \$197,908 for women even though expected cost barely moves (Figure 6): health at 65 matters for the tail more than for the mean.

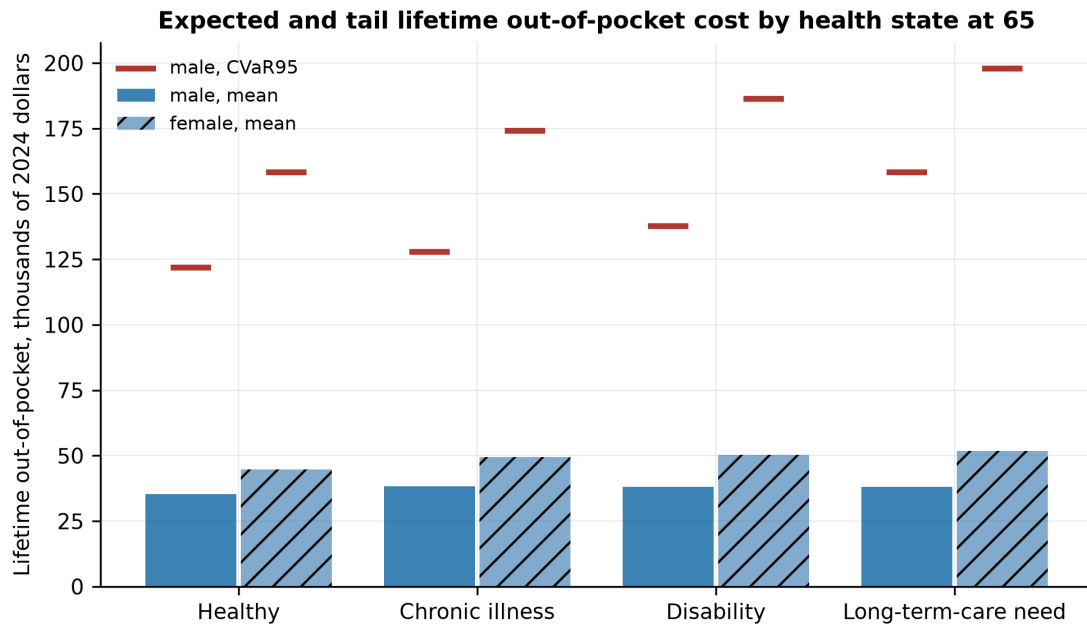


Figure 6. Expected lifetime out-of-pocket cost and CVaR95 by health state at 65.

6.6 Financing scenarios (RQ4)

Table 7 and Figure 7 report the scenarios. A pure provider payment cut (S1 at 0%) leaves household cost unchanged by construction, and its effect would come through access, which the model does not capture. Passing a quarter of the Part A shortfall to beneficiaries raises expected lifetime out-of-pocket cost by \$436 for men and \$521 for women; half raises it by \$872 and \$1,042; the whole shortfall by \$1,743 and \$2,084, and CVaR95 by \$2,403 and \$2,732. Concentrating the shortfall on person-years in long-term-care need (S2) leaves the mean change the same but raises CVaR95 by \$5,945 for men and \$6,380 for women, more than twice the effect of spreading it. Real cost growth at the Trustees' Part B rate with half the shortfall shifted (S3) raises the mean by \$9,428 and \$13,781 and CVaR95 by \$36,990 and \$55,219.

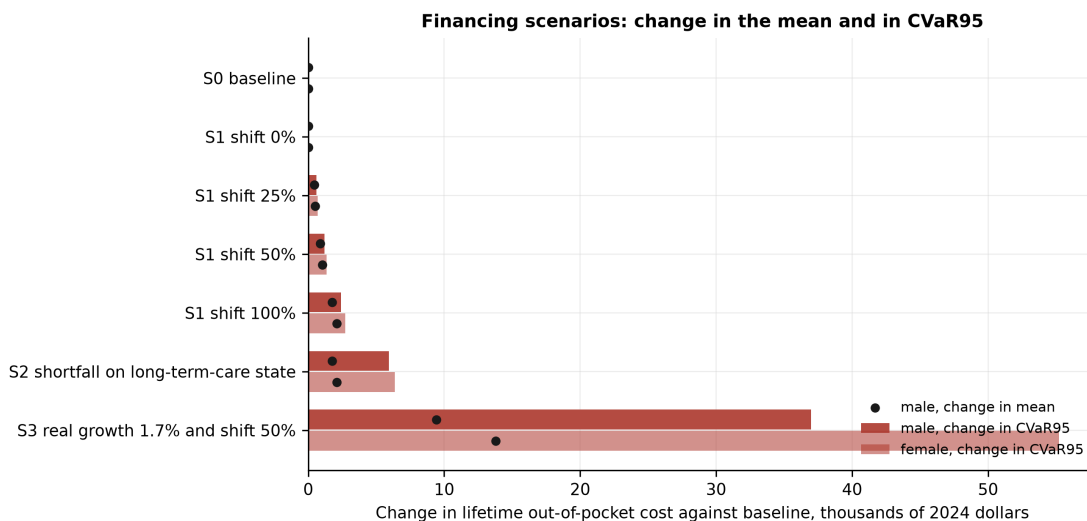


Figure 7. Change in the mean and in CVaR95 of lifetime out-of-pocket cost under the financing scenarios.

These effects are modest because Part A is a small part of Medicare spending in community settings: inpatient and home health are 17.6% of all-payer spending in the Cost Supplement, which excludes skilled nursing facilities. The shortfall scenarios are therefore lower bounds, and the growth scenario shows where the larger household risk lies.

Asset exhaustion is concentrated at low income (Table 7b). In the lowest income tertile, 19.0% of households have no positive assets at 65, and of the rest 32.2% of men and 36.9% of women spend their assets on out-of-pocket care costs, at a median age of 70 and 71. In the middle tertile the shares are 11.3% and 14.4%, and in the highest 3.6% and 4.8%. Shifting half the shortfall raises these shares by at most half a percentage point.

6.7 Robustness

Table 8 reports variants for the population mix with 40,000 lives per sex, so its baseline row differs from Table 5 by simulation error. The discount rate matters most for the tail: men's CVaR95 is \$146,755 at 2% and \$113,192 at 4%, against \$128,011 at 3%, and real cost growth at the Trustees' Part B rate raises it to \$163,364. A 25% uplift to Medicare cost in D and L, for the facility and hospice events the Cost Supplement omits, raises lifetime Medicare spending for men from \$153,106 to \$162,628; without the level scaling to MCBS it would be \$129,102. Without mortality calibration, life expectancy at 65 falls to 16.7 years for men and CVaR95 to \$122,640. Without the end-of-life step CVaR95 is \$121,323, so exit-interview costs add about 5% to the tail.

The refits change the state definitions or the sample, rebuild the entry mix, costs and calibration, and are compared with the sex-only model, whose CVaR95 for men is \$131,093. Defining L as two or more ADL limitations raises the share ever in L to 60% of men and 72% of women but moves men's expected out-of-pocket cost by \$134 and CVaR95 by \$116. Not counting nursing-home residence as L changes both by less than 1%. Restricting the panel to waves 9 to 16 (2008 to 2022) lowers expected out-of-pocket cost to \$35,392 for men and \$44,363 for women and CVaR95 to \$118,334 and \$155,362. In every variant CVaR95 stays between 3.3 and 3.6 times the mean for men and between 3.5 and 3.7 times for women.

7. Discussion

For Medicare policy. How the HI gap is closed matters less to the average household than to the few who need long-term care. A cut that falls on post-acute and long-term-care services concentrates the burden where out-of-pocket risk is already highest. The calculator reports these distributional effects, which aggregate solvency measures cannot show. The growth scenario is a reminder that the cost trend, not the 2033 date, drives most of the household risk.

For insurers and planners. Expected lifetime out-of-pocket cost is similar across health states at 65, but the tail is not, and it is driven by long-term-care paths and by the final year of life. Products that cover long-term-care need or end-of-life costs address the part of the distribution that households cannot self-insure through ordinary saving. The model's state intensities, calibrated to current mortality, can be used directly in pricing and reserving work with the covariate structure shown in Table 2, and the relatively flat lifetime Medicare cost across entry states supports the finding of Lubitz et al. (2003) with more recent data and a payer split.

For households. A typical 65-year-old should expect a present value of out-of-pocket care costs near \$38,000 for men and \$49,000 for women, plus about \$30,000 of Part B premiums, but should plan for the possibility of several times that. For low-income households with modest assets, the realistic outcome in the bad case is spending down to Medicaid.

Limitations. Five matter most. First, the MCBS Cost Supplement excludes facility, hospice and institutional events, so Medicare cost in D and L is understated, and the Part A share omits skilled nursing; the robustness uplift gives a sense of scale. Second, HRS out-of-pocket draws are independent across years within a state, so persistence beyond what the states capture is missing and the tail is understated. Third, the model is Markov in age with a log-linear age effect and no calendar trend; duration dependence in disability and mortality improvement after 2023 are not modeled, and calibration to a period table understates the longevity of a cohort turning 65 in 2026. Fourth, out-of-pocket spending excludes premiums other than Part B and, by design, nursing-home costs paid by Medicaid; the asset-exhaustion measure ignores income, investment returns and spouses. Fifth, the condition-count bridge between HRS and MCBS is approximate, and the level scaling

assumes the HRS state relativities carry over. Linked HRS-Medicare claims would address the first and last and are the natural next step.

8. Conclusion

A five-state model fitted to 24 years of HRS panel data, calibrated to current mortality and linked to payer-specific costs, shows that the expected lifetime cost of aging varies little with health at 65 while its tail varies a great deal, that the tail is made of long-term-care paths and the last year of life, and that a partial pass-through of Medicare's Part A shortfall matters far less to households than the long-run growth of costs. The released code rebuilds every number, and the calculator puts the results in front of the people who carry the risk.

Declarations

Data availability. The RAND HRS Longitudinal File and HRS exit data are available to registered users from the Health and Retirement Study; the MCBS Cost Supplement public use files from the Centers for Medicare & Medicaid Services; the life tables and Trustees Report are public. None is redistributed. The HRS (Health and Retirement Study) is sponsored by the National Institute on Aging (grant numbers NIA U01AG009740 and NIA R01AG073289) and is conducted by the University of Michigan.

Code availability. The seeded pipeline is at <https://github.com/tosin-babs/medicare-cost-of-aging>; `python/run_all.py` rebuilds every table and figure and `pytest tests` runs the checks. The public calculator is at <https://medicare-cost-of-aging.vercel.app>.

Competing interests. None declared.

Ethics. The analysis uses de-identified public-use survey data and did not require ethical approval.

AI-assistance disclosure. Generative AI (Claude, Anthropic) was used to assist with code development, code review and language editing. The author designed the study, specified all models and parameters, verified and interpreted all results, and takes full responsibility for the content. AI systems are not authors.

CRedit statement. Oluwatosin Dorcas Babalola: conceptualization, methodology, software, formal analysis, data curation, visualization, writing (original draft), writing (review and editing).

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Tables

Generated from `output/tables/*.csv` by `python/make_tables.py`. Dollar amounts are 2024 dollars; present values are discounted at 3% a year to age 65. HRS estimates use the respondent analysis weight, or the nursing-home resident weight for residents; MCBS estimates use the Cost Supplement weight with Fay-adjusted balanced repeated replication.

Table 1. The HRS panel, waves 4 to 16 (1998 to 2022), respondents aged 50 and over.

Quantity	Value
Respondents with at least one classified interview	40,354
Person-interviews with a classified state	237,944
Intervals between interviews	197,590
Deaths with an exact date	15,962
Mean interval length, years	2.0
Waves	4-16 (1998-2022)
Weighted share of person-interviews in Healthy, %	16.5
Weighted share of person-interviews in Chronic illness, %	68.6
Weighted share of person-interviews in Disability, %	10.8
Weighted share of person-interviews in Long-term-care need, %	4.1

Table 2. Transition intensities among health states.

Log-linear intensities fitted by maximum likelihood to interval-censored transitions. Reference: male, no college, non-Hispanic white, age 65. Hazard ratios with 95% intervals from the inverse observed information. Likelihood-ratio test of the education and race-ethnicity terms: 1,512.4 on 24 degrees of freedom.

Transition	Term	Estimate	SE	Hazard ratio	95% low	95% high
H to C	log rate at 65	-2.178	0.029			
H to C	age, per 10 years	0.123	0.017	1.13	1.09	1.17
H to C	female	-0.006	0.028	0.99	0.94	1.05
H to C	any college	-0.127	0.029	0.88	0.83	0.93
H to C	nonwhite or Hispanic	0.029	0.031	1.03	0.97	1.09
H to D	log rate at 65	-4.217	0.109			
H to D	age, per 10 years	0.758	0.049	2.13	1.94	2.35
H to D	female	-0.098	0.098	0.91	0.75	1.10
H to D	any college	-0.246	0.100	0.78	0.64	0.95

Transition	Term	Estimate	SE	Hazard ratio	95% low	95% high
H to D	nonwhite or Hispanic	0.513	0.102	1.67	1.37	2.04
H to X	log rate at 65	-5.416	0.204			
H to X	age, per 10 years	0.853	0.103	2.35	1.92	2.87
H to X	female	-0.858	0.230	0.42	0.27	0.67
H to X	any college	-0.781	0.239	0.46	0.29	0.73
H to X	nonwhite or Hispanic	0.332	0.224	1.39	0.90	2.16
C to D	log rate at 65	-2.900	0.022			
C to D	age, per 10 years	0.504	0.011	1.65	1.62	1.69
C to D	female	0.112	0.020	1.12	1.08	1.16
C to D	any college	-0.291	0.020	0.75	0.72	0.78
C to D	nonwhite or Hispanic	0.295	0.021	1.34	1.29	1.40
C to L	log rate at 65	-6.109	0.173			
C to L	age, per 10 years	-1.803	0.143	0.16	0.12	0.22
C to L	female	-0.228	0.121	0.80	0.63	1.01
C to L	any college	-0.907	0.134	0.40	0.31	0.53
C to L	nonwhite or Hispanic	0.854	0.130	2.35	1.82	3.03
C to X	log rate at 65	-4.149	0.038			
C to X	age, per 10 years	0.730	0.019	2.07	2.00	2.15
C to X	female	-0.563	0.036	0.57	0.53	0.61
C to X	any college	-0.222	0.036	0.80	0.75	0.86
C to X	nonwhite or Hispanic	0.104	0.041	1.11	1.02	1.20
D to H	log rate at 65	-4.573	0.105			
D to H	age, per 10 years	-0.451	0.055	0.64	0.57	0.71
D to H	female	-0.367	0.106	0.69	0.56	0.85
D to H	any college	0.224	0.109	1.25	1.01	1.55
D to H	nonwhite or Hispanic	0.187	0.110	1.21	0.97	1.50
D to C	log rate at 65	-1.126	0.023			
D to C	age, per 10 years	-0.086	0.011	0.92	0.90	0.94
D to C	female	0.018	0.023	1.02	0.97	1.07
D to C	any college	0.079	0.024	1.08	1.03	1.13
D to C	nonwhite or Hispanic	-0.034	0.024	0.97	0.92	1.01
D to L	log rate at 65	-1.924	0.035			
D to L	age, per 10 years	0.438	0.014	1.55	1.51	1.59
D to L	female	0.015	0.030	1.02	0.96	1.08
D to L	any college	-0.071	0.031	0.93	0.88	0.99

Transition	Term	Estimate	SE	Hazard ratio	95% low	95% high
D to L	nonwhite or Hispanic	0.165	0.031	1.18	1.11	1.25
D to X	log rate at 65	-2.870	0.076			
D to X	age, per 10 years	0.353	0.036	1.42	1.32	1.53
D to X	female	-0.438	0.085	0.65	0.55	0.76
D to X	any college	-0.363	0.096	0.70	0.58	0.84
D to X	nonwhite or Hispanic	-0.452	0.104	0.64	0.52	0.78
L to D	log rate at 65	-1.217	0.044			
L to D	age, per 10 years	-0.393	0.017	0.67	0.65	0.70
L to D	female	-0.087	0.041	0.92	0.85	0.99
L to D	any college	-0.035	0.043	0.97	0.89	1.05
L to D	nonwhite or Hispanic	0.101	0.041	1.11	1.02	1.20
L to X	log rate at 65	-1.788	0.039			
L to X	age, per 10 years	0.429	0.014	1.54	1.49	1.58
L to X	female	-0.385	0.031	0.68	0.64	0.72
L to X	any college	0.013	0.032	1.01	0.95	1.08
L to X	nonwhite or Hispanic	-0.277	0.034	0.76	0.71	0.81

Table 3. Mortality: the fitted model, the calibrated model and the 2023 US period life table.

Entry mix of HRS respondents aged 64 to 66. The multiplier scales death intensities from every live state and equals $\exp(c0 + c1 (age - 65)/10)$ at the calibrated values.

Sex	e65, fitted	e65, calibrated	e65, life table	Multiplier at 65	75	85	95
male	16.72	18.31	18.19	0.70	0.77	0.84	0.92
female	19.54	20.87	20.71	0.64	0.76	0.89	1.05

Sex	Age	qx, fitted	qx, calibrated	qx, life table
male	65	0.0198	0.0141	0.0162
male	70	0.0302	0.0230	0.0226
male	75	0.0459	0.0371	0.0335
male	80	0.0688	0.0587	0.0542
male	85	0.1009	0.0904	0.0922
male	90	0.1441	0.1348	0.1599
male	95	0.2002	0.1939	0.2563
female	65	0.0133	0.0086	0.0101
female	70	0.0207	0.0150	0.0148
female	75	0.0325	0.0260	0.0237
female	80	0.0505	0.0439	0.0405
female	85	0.0766	0.0719	0.0709

Sex	Age	qx, fitted	qx, calibrated	qx, life table
female	90	0.1128	0.1129	0.1285
female	95	0.1609	0.1696	0.2160

Table 4. Annual cost by health state and age band: Medicare and out of pocket.

Medicare: MCBS Cost Supplement cells (age by chronic-condition count) weighted by each HRS state's condition-count mix; it excludes facility, hospice and institutional events and is a lower bound for the disability and long-term-care states. Out of pocket: HRS, annualized from the two-year recall, premiums excluded.

State	Age	Share 0-1 conditions	2-3	4+	Medicare	Out of pocket, mean
Healthy	65-74	1.00	0.00	0.00	\$5,291	\$1,216
Healthy	75+	1.00	0.00	0.00	\$7,010	\$1,313
Chronic illness	65-74	0.28	0.55	0.17	\$10,000	\$2,032
Chronic illness	75+	0.21	0.57	0.22	\$12,029	\$2,256
Disability	65-74	0.13	0.48	0.39	\$12,658	\$2,730
Disability	75+	0.12	0.50	0.38	\$13,625	\$3,078
Long-term-care need	65-74	0.09	0.39	0.53	\$14,048	\$5,018
Long-term-care need	75+	0.10	0.41	0.49	\$14,527	\$10,364

Table 4b. The annual out-of-pocket distribution by state, age band and sex (HRS core interviews).

State	Age	Sex	n	Mean	p50	p90	p95	p99
Healthy	65-74	female	3,844	\$1,233	\$517	\$3,035	\$5,059	\$9,384
Healthy	65-74	male	3,412	\$1,198	\$374	\$2,766	\$4,592	\$11,467
Healthy	75+	female	1,877	\$1,227	\$450	\$3,093	\$4,892	\$11,922
Healthy	75+	male	1,575	\$1,425	\$435	\$3,198	\$4,856	\$16,385
Chronic illness	65-74	female	28,618	\$2,085	\$1,005	\$4,685	\$7,007	\$15,606
Chronic illness	65-74	male	22,432	\$1,972	\$909	\$4,489	\$7,019	\$15,699
Chronic illness	75+	female	24,613	\$2,316	\$1,046	\$4,991	\$7,486	\$19,126
Chronic illness	75+	male	17,988	\$2,174	\$1,030	\$5,021	\$7,508	\$16,070
Disability	65-74	female	4,222	\$2,814	\$1,093	\$6,156	\$9,277	\$23,505
Disability	65-74	male	2,917	\$2,623	\$1,017	\$6,824	\$9,191	\$22,779
Disability	75+	female	6,822	\$3,143	\$1,262	\$6,656	\$10,000	\$30,663
Disability	75+	male	4,038	\$2,965	\$1,292	\$6,454	\$10,629	\$24,592
Long-term-care need	65-74	female	1,901	\$5,073	\$991	\$10,930	\$19,606	\$83,499

State	Age	Sex	n	Mean	p50	p90	p95	p99
Long-term-care need	65-74	male	1,123	\$4,929	\$635	\$10,417	\$24,931	\$62,414
Long-term-care need	75+	female	6,177	\$10,829	\$1,763	\$33,612	\$56,427	\$122,600
Long-term-care need	75+	male	2,601	\$9,228	\$1,764	\$25,519	\$49,953	\$101,341

Table 4c. Out-of-pocket spending in the last year of life (HRS exit interviews).

Spending from the last core interview to death, pro-rated to twelve months where that window is longer. Unweighted; age band at death.

Age at death	Sex	n	Months, median	Mean	p50	p90	p95	p99	Whole window, mean
65-74	male	1,664	14	\$6,340	\$1,216	\$15,935	\$28,273	\$82,128	\$9,444
65-74	female	1,438	15	\$7,429	\$1,188	\$17,069	\$33,652	\$104,425	\$11,414
75+	male	4,986	15	\$9,913	\$1,650	\$24,853	\$50,660	\$125,678	\$14,483
75+	female	6,488	15	\$12,641	\$1,805	\$32,970	\$67,776	\$151,655	\$18,814

Table 5. Expected lifetime cost from age 65 by sex and health state at 65, present value at 3%.

100,000 simulated lives per row, calibrated mortality. Intervals are the 2.5th and 97.5th percentiles of the analytic expected value over 200 draws of the transition parameters. The analytic column is the forward recursion the simulation mean must match.

Sex	State at 65	e65	Ever LTC need	Medicare	low	high	Out of pocket	analytic	low	high	OOP median	Part B premium
male	Healthy	20.4	43.3%	\$140,295	\$138,892	\$141,466	\$35,373	\$35,401	\$35,083	\$35,691	\$28,827	\$31,993
male	Chronic illness	18.4	43.5%	\$157,627	\$156,219	\$158,575	\$38,205	\$38,128	\$37,809	\$38,409	\$31,997	\$29,399
male	Disability	16.0	55.3%	\$147,323	\$145,578	\$148,872	\$38,046	\$37,839	\$37,435	\$38,256	\$30,803	\$25,979
male	Long-term-care need	12.5	100.0%	\$126,204	\$123,775	\$128,520	\$38,065	\$38,131	\$37,307	\$38,773	\$27,543	\$20,917
male	Population mix	18.3	46.4%	\$153,316	\$151,948	\$154,224	\$37,660	\$37,703	\$37,398	\$38,002	\$31,282	\$29,224
female	Healthy	23.0	56.9%	\$157,174	\$156,126	\$158,443	\$44,736	\$44,575	\$44,249	\$44,978	\$35,152	\$34,855
female	Chronic illness	21.0	58.5%	\$176,905	\$176,074	\$178,149	\$49,450	\$49,349	\$49,048	\$49,797	\$39,378	\$32,442
female	Disability	18.8	68.0%	\$170,146	\$169,087	\$171,684	\$50,378	\$50,601	\$50,195	\$51,198	\$39,203	\$29,554
female	Long-term-care need	15.7	100.0%	\$153,454	\$151,444	\$155,697	\$51,752	\$52,147	\$51,387	\$52,990	\$38,550	\$25,170

Sex	State at 65	e65	Ever LTC need	Medicare	low	high	Out of pocket	analytic	low	high	OOP median	Part B premium
female	Population mix	20.8	60.6%	\$172,481	\$171,643	\$173,609	\$48,975	\$48,889	\$48,580	\$49,354	\$38,684	\$32,221

Table 5b. Expected years in each health state from 65, and the end-of-life component of out-of-pocket cost.

Sex	State at 65	e65	Healthy	Chronic illness	Disability	LTC need	End-of-life increment, PV
male	Healthy	20.4	7.4	9.6	2.0	1.4	\$2,691
male	Chronic illness	18.4	0.1	14.7	2.3	1.4	\$2,766
male	Disability	16.0	0.2	9.2	4.4	2.1	\$2,582
male	Long-term-care need	12.5	0.1	5.3	2.6	4.5	\$2,030
male	Population mix	18.3	1.2	13.2	2.5	1.6	\$2,663
female	Healthy	23.0	7.6	10.6	2.5	2.2	\$3,155
female	Chronic illness	21.0	0.1	15.6	2.9	2.4	\$3,170
female	Disability	18.8	0.2	10.2	5.1	3.3	\$2,997
female	Long-term-care need	15.7	0.1	6.3	3.2	6.0	\$2,549
female	Population mix	20.8	1.2	14.0	3.1	2.6	\$3,072

Table 5c. Lifetime cost and tail risk by household income tertile at 65, population mix.

Income tertiles within interview wave and age group. Out-of-pocket draws come from the same tertile at every age. The transition model has no income term, so life-expectancy differences by income are understated.

Sex	Income	Healthy at 65	LTC need at 65	Any college	e65	Ever LTC need	Medicare	Out of pocket	OOP median	CVaR95	CVaR99
male	low	12.0%	6.8%	37%	17.4	51.1%	\$144,095	\$30,387	\$24,262	\$112,297	\$172,784
female	low	9.9%	8.0%	36%	20.1	64.7%	\$164,936	\$42,713	\$32,503	\$163,939	\$270,549
male	middle	13.5%	2.3%	53%	18.2	45.8%	\$153,486	\$36,836	\$30,849	\$123,238	\$184,223
female	middle	13.9%	2.3%	53%	21.0	59.1%	\$173,588	\$52,606	\$41,465	\$177,706	\$254,695
male	high	17.5%	0.8%	76%	19.1	43.6%	\$161,087	\$44,277	\$36,358	\$151,791	\$226,115
female	high	21.2%	0.8%	73%	21.8	57.4%	\$179,788	\$59,647	\$46,831	\$214,186	\$352,606

Table 6. The tail of lifetime out-of-pocket cost.

VaR is the quantile of the present value of lifetime out-of-pocket cost; CVaR is the mean beyond it. LTC columns are the percentages of lives that ever entered the long-term-care-need state, in the tail and overall. The end-of-life share is the part of the tail's present value that comes from the final year's excess over an ordinary year.

Sex	State at 65	Level	VaR	CVaR	CVaR / mean	LTC in tail	LTC overall	End-of-life share of tail
male	Healthy	95%	\$86,637	\$121,865	3.4	72.9%	43.3%	21.4%
male	Healthy	99%	\$143,643	\$185,781	5.3	66.6%	43.3%	22.0%
male	Chronic illness	95%	\$92,141	\$127,813	3.3	75.0%	43.5%	19.9%

Sex	State at 65	Level	VaR	CVaR	CVaR / mean	LTC in tail	LTC overall	End-of-life share of tail
male	Chronic illness	99%	\$146,630	\$193,398	5.1	74.3%	43.5%	21.3%
male	Disability	95%	\$99,727	\$137,762	3.6	84.6%	55.3%	18.1%
male	Disability	99%	\$159,174	\$203,938	5.4	82.8%	55.3%	19.9%
male	Long-term-care need	95%	\$116,010	\$158,274	4.2	100.0%	100.0%	12.6%
male	Long-term-care need	99%	\$184,878	\$225,998	5.9	100.0%	100.0%	13.8%
male	Population mix	95%	\$92,761	\$127,541	3.4	77.8%	46.4%	19.8%
male	Population mix	99%	\$148,529	\$188,051	5.0	72.9%	46.4%	23.1%
female	Healthy	95%	\$110,499	\$158,291	3.5	85.9%	56.9%	19.1%
female	Healthy	99%	\$181,042	\$253,353	5.7	81.6%	56.9%	22.1%
female	Chronic illness	95%	\$121,535	\$173,995	3.5	85.9%	58.5%	16.6%
female	Chronic illness	99%	\$200,697	\$277,558	5.6	80.1%	58.5%	18.4%
female	Disability	95%	\$131,855	\$186,309	3.7	91.2%	68.0%	13.4%
female	Disability	99%	\$208,306	\$296,705	5.9	88.5%	68.0%	15.9%
female	Long-term-care need	95%	\$146,020	\$197,908	3.8	100.0%	100.0%	11.0%
female	Long-term-care need	99%	\$222,409	\$294,386	5.7	100.0%	100.0%	14.0%
female	Population mix	95%	\$122,555	\$174,866	3.6	88.3%	60.6%	16.1%
female	Population mix	99%	\$198,738	\$282,008	5.8	84.0%	60.6%	18.4%

Table 7. Financing scenarios: lifetime out-of-pocket cost for a cohort turning 65 in 2026, population mix.

S1: from 2033, Part A pays 89% of scheduled benefits and the stated share of the shortfall falls on beneficiaries. S2: the whole cohort shortfall falls on person-years in long-term-care need. S3: S1 at 50% with real cost growth at the Trustees' Part B per-beneficiary rate.

Sex	Scenario	Mean	VaR95	CVaR95	CVaR99	Change in mean	Change in CVaR95
male	S0 baseline	\$37,660	\$92,761	\$127,541	\$188,051	\$0	\$0
male	S1 Part A payable 89% from 2033, shift 0%	\$37,660	\$92,761	\$127,541	\$188,051	\$0	\$0
male	S1 Part A payable 89% from 2033, shift 25%	\$38,096	\$93,416	\$128,139	\$188,622	\$436	\$598
male	S1 Part A payable 89%	\$38,532	\$94,123	\$128,739	\$189,195	\$872	\$1,198

Sex	Scenario	Mean	VaR95	CVaR95	CVaR99	Change in mean	Change in CVaR95
	from 2033, shift 50%						
male	S1 Part A payable 89% from 2033, shift 100%	\$39,404	\$95,302	\$129,944	\$190,346	\$1,743	\$2,403
male	S2 shortfall on long-term-care state	\$39,404	\$97,468	\$133,487	\$194,797	\$1,743	\$5,945
male	S3 real growth 1.7% and shift 50%	\$47,089	\$120,209	\$164,531	\$239,987	\$9,428	\$36,990
female	S0 baseline	\$48,975	\$122,555	\$174,866	\$282,008	\$0	\$0
female	S1 Part A payable 89% from 2033, shift 0%	\$48,975	\$122,555	\$174,866	\$282,008	\$0	\$0
female	S1 Part A payable 89% from 2033, shift 25%	\$49,496	\$123,198	\$175,547	\$282,682	\$521	\$681
female	S1 Part A payable 89% from 2033, shift 50%	\$50,017	\$123,847	\$176,230	\$283,357	\$1,042	\$1,363
female	S1 Part A payable 89% from 2033, shift 100%	\$51,059	\$125,192	\$177,598	\$284,713	\$2,084	\$2,732
female	S2 shortfall on long-term-care state	\$51,059	\$127,870	\$181,247	\$288,305	\$2,084	\$6,380
female	S3 real growth 1.7% and shift 50%	\$62,756	\$161,692	\$230,086	\$367,038	\$13,781	\$55,219

Table 7b. Exhausting household assets at 65 through out-of-pocket cost, by household income tertile at 65.

Undiscounted cumulative out-of-pocket cost against household net assets at 65 in 2024 dollars, with no asset returns or income. The exhaustion share is among households with positive assets.

Sex	Income	Scenario	Median assets	No positive assets	Exhaust assets	Median age at exhaustion
male	low	S0 baseline	\$66,192	19.0%	32.2%	70
male	low	S1 shift 50%	\$66,192	19.0%	32.6%	70
male	middle	S0 baseline	\$304,270	4.3%	11.3%	75
male	middle	S1 shift 50%	\$304,270	4.3%	11.6%	75
male	high	S0 baseline	\$817,677	1.7%	3.6%	77
male	high	S1 shift 50%	\$817,677	1.7%	3.7%	77
female	low	S0 baseline	\$66,388	18.8%	36.9%	71

Sex	Income	Scenario	Median assets	No positive assets	Exhaust assets	Median age at exhaustion
female	low	S1 shift 50%	\$66,388	18.8%	37.4%	71
female	middle	S0 baseline	\$305,424	4.3%	14.4%	77
female	middle	S1 shift 50%	\$305,424	4.3%	14.6%	77
female	high	S0 baseline	\$816,875	1.7%	4.8%	79
female	high	S1 shift 50%	\$816,875	1.7%	4.9%	79

Table 8. Robustness: each row changes one decision.

Population mix at 65, calibrated mortality unless stated; 40,000 lives per sex. Refits use the sex-only covariate set and rebuild the entry mix, state costs and calibration from the variant panel; compare them with the sex-only row.

Variant	e65 M	e65 F	Ever LTC M	Ever LTC F	Medicare M	Medicare F	OOP M	OOP F	CVaR95 M	CVaR95 F
baseline	18.3	20.9	46%	61%	\$153,106	\$172,744	\$37,606	\$49,194	\$128,011	\$173,690
discount 2%	18.3	20.9	46%	61%	\$169,324	\$192,785	\$42,087	\$55,911	\$146,755	\$201,653
discount 4%	18.3	20.9	46%	61%	\$139,336	\$155,907	\$33,848	\$43,644	\$113,192	\$151,794
Trustees real growth 1.7%	18.3	20.9	46%	61%	\$182,867	\$209,680	\$45,869	\$61,657	\$163,364	\$226,780
D and L Medicare +25%	18.3	20.9	46%	61%	\$162,628	\$185,857	\$37,606	\$49,194	\$128,011	\$173,690
Medicare not scaled to MCBS	18.3	20.9	46%	61%	\$129,102	\$145,787	\$37,606	\$49,194	\$128,011	\$173,690
mortality not calibrated	16.7	19.5	41%	56%	\$141,280	\$161,842	\$34,435	\$45,169	\$122,640	\$162,970
no end-of-life step	18.4	20.9	46%	61%	\$153,446	\$172,474	\$35,162	\$45,957	\$121,323	\$165,063
sex-only model	18.3	20.9	49%	63%	\$153,547	\$173,164	\$38,237	\$50,215	\$131,093	\$177,181
L at 2+ ADLs (refit)	18.3	20.8	60%	72%	\$153,730	\$172,475	\$38,371	\$49,448	\$130,977	\$171,743
nursing home not L (refit)	18.3	20.8	46%	60%	\$153,772	\$173,129	\$38,405	\$49,987	\$130,254	\$175,784
waves 9-16 only (refit)	18.3	20.8	48%	62%	\$153,036	\$172,062	\$35,392	\$44,363	\$118,334	\$155,362

Appendix tables

Table A1. Observed transitions between consecutive interviews, and deaths.

Counts of intervals by state at the start (rows) and at the end (columns). Deaths are dated to the month.

From	Healthy	Chronic illness	Disability	Long-term-care need	Dead
Healthy	24,501	5,808	768	232	423
Chronic illness	0	119,618	11,702	3,296	7,384
Disability	332	8,537	9,149	3,877	3,362
Long-term-care need	27	950	2,248	6,545	4,793

Table A2. One-year transition probabilities from the fitted model, reference covariates, before calibration.

Sex	Age	From	Healthy	Chronic	Disability	LTC need	Dead
male	65	Healthy	0.875	0.105	0.014	0.001	0.006
male	65	Chronic illness	0.000	0.935	0.042	0.005	0.017
male	65	Disability	0.007	0.242	0.602	0.092	0.056
male	65	Long-term-care need	0.001	0.033	0.178	0.644	0.144
male	75	Healthy	0.842	0.115	0.027	0.003	0.013
male	75	Chronic illness	0.000	0.891	0.066	0.008	0.036
male	75	Disability	0.004	0.209	0.562	0.137	0.088
male	75	Long-term-care need	0.000	0.020	0.116	0.646	0.218
male	85	Healthy	0.786	0.123	0.051	0.008	0.032
male	85	Chronic illness	0.000	0.813	0.097	0.017	0.073
male	85	Disability	0.003	0.171	0.496	0.192	0.138
male	85	Long-term-care need	0.000	0.011	0.071	0.599	0.319
female	65	Healthy	0.879	0.104	0.013	0.001	0.003
female	65	Chronic illness	0.000	0.937	0.048	0.005	0.010
female	65	Disability	0.005	0.248	0.612	0.098	0.037
female	65	Long-term-care need	0.001	0.032	0.171	0.695	0.101
female	75	Healthy	0.850	0.115	0.026	0.003	0.006
female	75	Chronic illness	0.000	0.895	0.075	0.009	0.021
female	75	Disability	0.003	0.215	0.575	0.148	0.058
female	75	Long-term-care need	0.000	0.019	0.113	0.713	0.155
female	85	Healthy	0.803	0.124	0.049	0.009	0.015
female	85	Chronic illness	0.000	0.825	0.111	0.021	0.043
female	85	Disability	0.002	0.178	0.513	0.213	0.094
female	85	Long-term-care need	0.000	0.011	0.071	0.687	0.231

Table A3. State prevalence by age band: the weighted HRS cross-section against the calibrated model (percent of survivors).

Sex	Age	Healthy, HRS	Healthy, model	Chronic illness, HRS	Chronic illness, model	Disability, HRS	Disability, model	Long-term-care need, HRS	Long-term-care need, model
female	65-69	11.9%	11.8%	74.1%	73.1%	9.7%	10.5%	4.3%	4.7%
female	70-74	8.7%	6.9%	75.6%	72.9%	10.5%	13.1%	5.2%	7.1%
female	75-79	6.4%	4.1%	72.1%	69.6%	13.7%	15.4%	7.7%	10.9%
female	80-84	4.8%	2.5%	66.0%	64.4%	16.6%	17.3%	12.6%	15.9%
female	85-89	3.3%	1.6%	55.7%	58.3%	20.5%	18.6%	20.6%	21.6%
female	90-99	2.0%	1.0%	39.9%	50.4%	23.6%	19.5%	34.4%	29.1%
male	65-69	12.7%	12.1%	75.7%	74.8%	8.5%	9.6%	3.1%	3.5%
male	70-74	8.8%	7.3%	77.3%	75.6%	10.1%	11.7%	3.8%	5.4%
male	75-79	6.2%	4.4%	76.1%	73.5%	12.2%	13.9%	5.6%	8.2%
male	80-84	5.2%	2.8%	70.0%	69.6%	16.0%	15.9%	8.8%	11.7%
male	85-89	5.4%	1.9%	62.9%	64.7%	17.7%	17.5%	14.0%	15.9%
male	90-99	4.8%	1.3%	51.6%	58.1%	20.3%	19.1%	23.3%	21.5%

Table A4. Medicare spending per person-year by age band: the model cohort against the MCBS Cost Supplement.

The model applies the MCBS cells to the simulated cohort's state and condition mix; a gap reflects the difference between that mix and the MCBS population's.

Age	Model	MCBS	MCBS SE	Model / MCBS
65-74	\$10,051	\$9,920	\$357	1.013
75+	\$12,544	\$12,343	\$349	1.016

Table A5. Life expectancy at 65 by covariate profile and health state at 65, fitted model before calibration.

Sex	Any college	Nonwhite or Hispanic	Healthy	Chronic illness	Disability	LTC need
female	no	no	20.6	18.5	16.0	12.1
female	no	yes	20.0	18.2	16.5	13.7
female	yes	no	23.2	20.9	18.4	13.4
female	yes	yes	22.3	20.3	18.5	14.9
male	no	no	17.7	15.5	13.0	9.2
male	no	yes	16.9	15.2	13.5	10.6
male	yes	no	20.3	17.9	15.3	10.4
male	yes	yes	19.3	17.2	15.4	11.7

Table A6. MCBS Cost Supplement: annual spending per beneficiary by payer, 2019 and 2021 to 2023 pooled.

Medicare includes fee-for-service and Medicare Advantage payments. Standard errors from Fay-adjusted BRR with 100 replicates.

Cell	n	In MA	Medicare	SE	Medicaid	Out of pocket	SE	All payers	Part A share
All	29,144	44.2%	\$11,802	\$278	\$572	\$2,717	\$46	\$17,406	17.6%

Cell	n	In MA	Medicare	SE	Medicaid	Out of pocket	SE	All payers	Part A share
65-74	10,346	42.4%	\$9,920	\$357	\$315	\$2,510	\$60	\$14,974	16.2%
75 and over	14,033	46.3%	\$12,343	\$349	\$386	\$3,228	\$69	\$18,330	18.6%
65-74, 0-1 chronic conditions	2,239	34.9%	\$4,418	\$493	\$42	\$1,624	\$65	\$7,264	13.4%
65-74, 2-3 chronic conditions	4,282	42.2%	\$8,217	\$445	\$118	\$2,389	\$89	\$12,909	16.1%
65-74, 4+ chronic conditions	3,824	47.6%	\$15,499	\$720	\$725	\$3,229	\$107	\$22,414	16.7%
75 and over, 0-1 chronic conditions	1,791	43.9%	\$5,976	\$405	\$93	\$2,445	\$144	\$9,718	16.7%
75 and over, 2-3 chronic conditions	5,402	44.9%	\$9,648	\$402	\$122	\$2,868	\$81	\$14,575	16.7%
75 and over, 4+ chronic conditions	6,840	48.0%	\$15,930	\$542	\$656	\$3,690	\$98	\$23,267	19.7%
65+, 0-1 chronic conditions, Income \$25,000 or less	824	40.2%	\$5,516	\$835	\$265	\$818	\$69	\$7,137	13.7%
65+, 0-1 chronic conditions, Income over \$25,000	3,206	36.6%	\$4,685	\$399	\$10	\$2,067	\$78	\$8,090	14.6%
65+, 2-3 chronic conditions, Income \$25,000 or less	2,239	58.8%	\$9,743	\$998	\$519	\$1,336	\$93	\$12,591	17.1%
65+, 2-3 chronic conditions, Income over \$25,000	7,445	39.4%	\$8,520	\$372	\$22	\$2,874	\$81	\$13,776	16.2%
65+, 4+ chronic conditions, Income \$25,000 or less	3,377	61.8%	\$16,001	\$702	\$2,182	\$1,872	\$104	\$21,208	19.5%

Cell	n	In MA	Medicare	SE	Medicaid	Out of pocket	SE	All payers	Part A share
65+, 4+ chronic conditions, Income over \$25,000	7,287	42.2%	\$15,594	\$626	\$95	\$4,088	\$100	\$23,482	17.8%

Table A7. 2023 US period life table, selected ages (NCHS).

Age	qx, male	ex, male	qx, female	ex, female
65	0.01620	18.19	0.01013	20.71
70	0.02264	14.74	0.01476	16.82
75	0.03352	11.48	0.02371	13.15
80	0.05416	8.53	0.04049	9.85
85	0.09216	6.06	0.07093	7.04
90	0.15990	4.14	0.12854	4.82
95	0.25628	2.83	0.21598	3.26
100	1.00000	2.01	1.00000	2.26